Where Cross-Question Structure in LLM Errors Comes From: Shared Seeds, Shared Prompts, and Answer-Letter Preference

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Abstract

Disagreement among a language model's samples, or among the models of an ensemble, is read as epistemic uncertainty: models that lack some knowledge should fail on the same groups of questions. We test when such structure across the questions of a correctness matrix reflects shared knowledge, by counting the eigenvalues of the question correlation matrix beyond those of independent answers or, across models, of a fitted item response model with one ability per model. Within one model, where only the sampling and prompting pipeline can couple two questions, 12 runs with independent draws show no detectable structure, while a sampling seed or a few-shot prompt shared across questions creates it. Across 24 greedily decoded models, the largest structure beyond ability is answer-letter preference: when the options are rotated it follows mostly the displayed letter, not the question, and it comes mostly from one model. When a question counts as correct only if it is correct in all four rotations, the second eigenvalue lies below the 99th percentile of valid one-ability models. In post hoc simulations from the fitted model, a second ability as strong as the first on the third of the questions from STEM subjects would usually lift it above that percentile, but one half as strong would not. A marginal excess, detected only jointly over the second to fourth eigenvalues, comes mostly from two pairs of near-duplicate models and partly from the letter-biased model. Options should be rotated before an ensemble's error structure is read as epistemic.

1. Introduction

Self-consistency samples a language model K times on each question and reads disagreement among the samples as uncertainty about the answer (Wang et al., 2023; Manakul et al., 2023; Farquhar et al., 2024). Disagreement among the members of a deep ensemble (Lakshminarayanan et al., 2017) is read as epistemic uncertainty, from missing knowledge, rather than aleatoric uncertainty, the noise that remains when the model is known (Kendall & Gal, 2017; Hüllermeier & Waegeman, 2021). Models that lack some knowledge fail on the questions that need it, so their errors should be related across questions: models often choose the same wrong answer (Kim et al., 2025), and a few latent factors summarize many models' scores (Burnell et al., 2023; Ruan et al., 2024; Maimon et al., 2026). Scores that add cross-model disagreement to self-consistency as an epistemic term (Hamidieh et al., 2026) read disagreement as missing knowledge. A per-question score cannot show why models disagree; the pattern across questions can. Disagreement from missing knowledge should follow question content, and disagreement from a letter preference should follow the displayed letter. We ask when *cross-question structure*, the tendency of questions to be answered right or wrong together, reflects shared knowledge, and when it comes from something else.

We call the kth samples of all questions *pass* k; like one ensemble member, a pass answers every question once. Within one model, a knowledge gap lowers the rate at which the passes answer a question correctly, but it does not make two questions right or wrong together from pass to pass. Given the model, only the pipeline (seeding, batching, prompt construction) can couple them: one seed in a batched call gives every prompt of a pass the same random numbers, and the standard 5-shot MMLU prompt places one set of exemplars before every question of a subject (Hendrycks et al., 2021). Within-model runs are therefore controls in which shared knowledge cannot create structure. Across models, ability couples the questions, and so can habits unrelated to knowledge: a model that prefers one answer letter is right more often on the questions whose correct option carries that letter (Zheng et al., 2024; Pezeshkpour & Hruschka, 2024).

We measure cross-question structure by the eigenvalues of the question correlation matrix, against independent answers (Marčenko & Pastur, 1967; Johnstone, 2001) within a model and against one ability per model, as in item response models (Rasch, 1960; Lord & Novick, 1968), across models

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(App. A). A group of questions answered right or wrong together lifts an eigenvalue above its reference; we call the corresponding eigenvector a *direction*. Across models the first direction is ability; the question is what lies beyond it. Runs whose samples are drawn independently show none; a shared seed or prompt creates one (Sec. 3.1). Across 24 models, the second direction is letter preference, which we identify by rotating the options (Sec. 3.2). Scoring a question correct only if it is correct in every rotation, the CircularEval rule of Liu et al. (2024), removes a letter preference that acts on guesses and leaves a marginal excess carried mostly by near-duplicate models (Sec. 3.3). We test the premise that cross-question structure in correctness reflects shared knowledge; we do not evaluate an uncertainty score. Spectral dimensionality measures have been applied to item-level benchmark matrices (Sha & Zhao, 2026), and letter preference and the pull of few-shot exemplars are known for single models (Zheng et al., 2024; Pezeshkpour & Hruschka, 2024; Zhao et al., 2021; Min et al., 2022; Ding, 2026) (App. A). We show that a correctness matrix, which records no letter, still carries one model's letter preference as a direction of the across-model matrix.

2. A spectral test for cross-question structure

Entry c_{ki} of the correctness matrix $\mathbf{C}$ is 1 when row k answers question i correctly and 0 otherwise. Within one model, row k is pass k; across models, each row is one model. We keep the questions that vary: within a model, the N_b *borderline* questions whose pass rate f_i, the fraction of passes that answer question i correctly, satisfies $\varepsilon < f_i < 1 - \varepsilon$, $\varepsilon = 0.05$ (App. C varies it from 0.01 to 0.20); across models, the *split* questions, which some models answer correctly and others do not. Let $\mathbf{R}$ be the Pearson correlation matrix of the kept columns over the rows. If the questions are independent, the eigenvalues of $\mathbf{R}$ follow the Marchenko–Pastur (MP) law as rows and columns grow in proportion; with γ the ratio of columns to rows, its upper edge is $\lambda_+ = (1 + \sqrt{\gamma})^2$ (Marčenko & Pastur, 1967). A direction gives each question's loading, and a row's *score* on it is the projection of the row after each column is standardized.

Within one model. We compare the largest eigenvalue $\lambda_{\max}$ with λ_+. The Tracy–Widom statistic z measures the distance of $\lambda_{\max}$ above λ_+ in units of its typical fluctuation (Johnstone, 2001); we calibrate z against 1000 matrices of independent Bernoulli entries at the observed pass rates, with the filter re-applied (App. C). The MP count is the number of eigenvalues above λ_+.

Across models. Here the reference is a one-ability item response model fitted to the data (fitting details: App. F). In the two-parameter logistic (2PL) model (Lord & Novick, 1968), model m answers question i correctly with probability $\sigma(a_i(\theta_m - b_i))$, where σ is the logistic function, θ_m the model's ability, and b_i and a_i the question's difficulty and slope. Rasch sets every slope equal (Rasch, 1960), and the three-parameter (3PL) model adds a fixed guessing floor of 0.25. Each model answers every question in four *rotations*: rotations 1, 2 and 3 move every option one, two or three letters later, cyclically, and rotation 0 is the original. In the uniform know-or-guess (KG-u) null, a model knows a question in all four rotations at once, with the 2PL probability, or else guesses uniformly among the four letters in each rotation. A one-ability model can by itself lift a second eigenvalue above λ_+: with 0/1 answers, one ability can produce a second component related to question difficulty, the *difficulty factor* of binary items (Ferguson, 1941; McDonald & Ahlawat, 1974). We therefore compare each eigenvalue λ_j with the jth eigenvalue of 1000 matrices simulated from the fitted null; rank j *exceeds* the null when λ_j is above the null's 99th percentile. A null counts only if it is *valid*: the observed λ_1 lies between the 1st and 99th percentiles of its draws. λ_1 measures how tightly ability ties the answers together, and a null that misses it misjudges the size of the eigenvalues after it (App. F). An excess can spread over several ranks, so a *family-wise test* uses the statistic M, the largest of the percentile ranks of λ_2, λ_3 and λ_4 among the null draws; p_M is the share of null draws whose own M, computed against the other draws, is at least as large. We fixed this test, the validity rule and the *sibling groups* (versions or fine-tunes of one base model at one size: Phi-3/3.5-mini, Llama-3/3.1-8B, Qwen2/2.5-7B, and Mistral-7B v0.2/v0.3 with Zephyr-7B) before running the tests, but after a similar statistic and the model-pair correlations had been computed on the same data. The removals of single models or model pairs, and the size, power and difficulty analyses of Sec. 3.3, are post hoc.

Runs. Within one model (App. B), an *independent-draw run* gives every (question, pass) pair its own seed, as self-consistency does; in a *shared-seed run* all questions of a pass use one seed; in a *shared-prompt run* each pass draws five MMLU development questions, with their answer letters, and places them as exemplars before every question of the pass. Ten independent-draw runs use temperature 1: five models on 500 MMLU test questions (Hendrycks et al., 2021) (ten from each of 50 subjects), three models on HellaSwag (Zellers et al., 2019), and Qwen2.5-7B on GSM8K (Cobbe et al., 2021) at two output lengths. Two more use Qwen2.5-7B on MMLU at temperatures 0.5 and 1.5. These twelve runs have $K = 30$ to 100 passes (Table A1). One shared-seed and two shared-prompt runs use MMLU with $K = 100$. Across models, 24 models of 0.5B to 22B parameters from ten families answer the 500 MMLU questions by greedy decoding (always the most probable answer) in all four rotations.

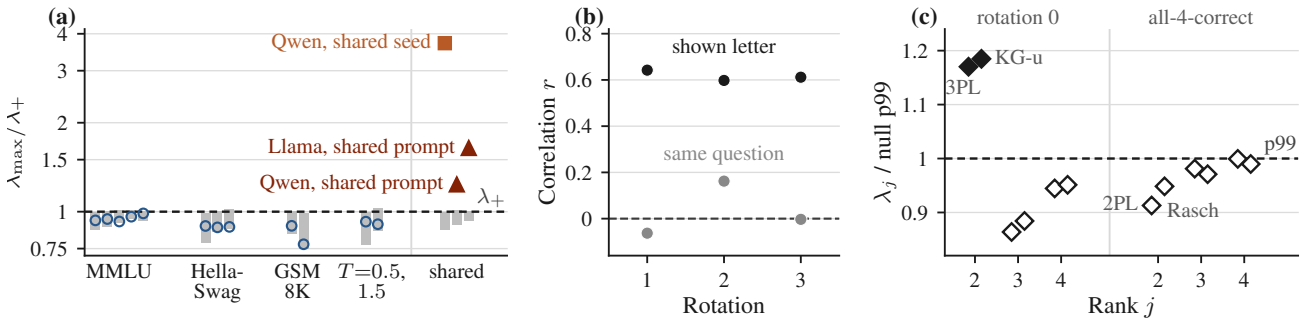


Figure 1. Within one model (a), across models (b), and after rotation (c). **(a)** $\lambda_{\max}/\lambda_+$ (log axis) of each within-model run; grey bar: 5–95% of its null; circles: independent draws. **(b)** Correlation of the rotated eigenvector-2 loadings with the rotation-0 mean loading of the shown letter, and with the question's own rotation-0 loading. **(c)** λ_2–λ_4 of the 24-model matrix in rotation 0 and all-4-correct (Sec. 3.3), divided by the 99th percentile of two valid nulls in 1000 draws (left marker: 3PL or 2PL; right: KG-u or Rasch); filled: λ_j above that 99th percentile.

3. Results

3.1. Within one model: structure only from a shared component

In all 12 independent-draw runs the largest eigenvalue lies below λ_+ (Fig. 1a), every z lies between the 10th and 84th percentiles of its null, and the MP count stays zero at six borderline thresholds (App. C). To measure power, we plant a factor shared by the borderline questions of a pass, with latent correlation ρ between questions; their 0/1 answers then correlate at about $\rho/2$ at these pass rates (App. D). At $\rho = 0.05$ the MP count detects it in every simulated matrix at Llama-3-8B's $N_b = 173$ and in 47% at Qwen2.5-7B's $N_b = 43$. A cluster of nine questions at $\rho = 0.2$ is detected in at most 23%. Weak structure in runs with few borderline questions, and small clusters, are therefore not excluded.

In the shared-seed and shared-prompt runs, z exceeds every draw of its null and two to four eigenvalues exceed λ_+; we trace the top direction of each to its source (App. E). The sampler of vLLM (Gumbel-max sampling) divides each token's probability by a random number and emits the token with the largest ratio, and a seed in a batched call's sampling parameters gives every question of a pass the same random numbers, so the letter favored in one question is favored in all. In Qwen2.5-7B's shared-seed run, gold letter (the letter of the correct option) explains a share $\eta^2 = 0.73$ of the variance of the top eigenvector's loadings (permutation $p = 0.0002$). With a shared prompt, Llama-3-8B copies the exemplars' answer letters ($\eta^2 = 0.53$, $p = 0.0002$). Qwen2.5-7B's shared-prompt direction does not group by gold letter ($p = 0.094$), but two passes whose randomly drawn exemplar sets include the same question answer more alike than other pairs of passes ($p = 0.0005$), so the exemplar set couples its answers. A shared seed leaves each question's answer distribution unchanged, so per-question self-consistency is unaffected; but the passes are no longer independent members of an ensemble.

3.2. Across models: the second direction is letter preference

In rotation 0 the four largest eigenvalues of the 24-model matrix (460 split questions, $\lambda_+ = 28.92$) are 85.9, 49.9, 29.7 and 27.3. The models' scores on eigenvector 1 correlate with accuracy at $r = 0.94$: it is the ability direction. In rotation 0 a model can guess right, and only the nulls that allow guessing are valid: the 2PL and Rasch nulls put the observed λ_1 above all their draws, and the 3PL and KG-u nulls do not. Of ranks 2 to 4, only rank 2 exceeds both (Fig. 1c), and the family-wise test rejects both ($p_M = 0.002$).

The loadings of eigenvector 2 group by gold letter ($\eta^2 = 0.48$, $p = 0.0002$), and the models' scores on it correlate with their share of D answers at $r = -0.90$ and of A answers at $r = 0.76$. Within one rotation a question's gold letter is tied to its content, so this grouping cannot separate letter from content; rotating the options can. In rotations 1–3 the models' scores on eigenvector 2 stay close to those of rotation 0 ($|r|$ from 0.95 to 0.98). We correlate each question's loading with its own rotation-0 loading, and with the rotation-0 mean loading of the questions whose gold letter is the one it now shows. A direction of question content would follow its own loading; this one follows the shown letter ($r = 0.60$ to 0.64, against -0.06 to 0.16; Fig. 1b), although a small part that follows content is not excluded. A *soft-preference* null, one ability plus a fitted letter preference that acts on every choice, is valid here and not rejected by the family-wise test ($p_M = 0.079$), but it places λ_2 at $p = 0.010$ against 200 datasets simulated from its fit (App. F). The evidence that the direction is letter preference is the rotation test, not this null.

The direction comes mostly from one model, Llama-3.2-1B, which picks A nine times in ten when it guesses (estimated by a know-or-guess model fitted to all four rotations). Removing it lowers λ_2 from 49.9 to 35.0, below the 99th percentile of the 3PL null refitted to the other 23 models (40.1); removing any other single model leaves λ_2 between 47.1 and 52.4. Without Llama-3.2-1B a weaker excess remains at rank 4, and the family-wise test still rejects the 3PL null ($p_M = 0.004$).

3.3. After rotation: a marginal excess, mostly from two model pairs

We count a model correct on a question only if it is correct in all four rotations. The gold option appears once at each letter over the rotations, so a model that only guesses by letter is correct in all four with probability $\prod_L \pi_m(L) \leq 1/256$, where $\pi_m(L)$ is its probability of guessing letter L; this is the same for every question. This *all-4-correct* score removes a letter preference that acts on guesses; whether one that also acts on partly known choices survives cannot be tested, as the soft-preference null is invalid here (App. F). The all-4-correct matrix (391 split questions, $\lambda_+ = 25.36$) has eigenvalues 92.1, 29.0, 23.3 and 20.8.

Rank 2. With guessing nearly excluded, the validity pattern of rotation 0 reverses: the 2PL and Rasch nulls are valid, and the 3PL and KG-u nulls are not. The 2PL and Rasch 99th percentiles of λ_2, 31.9 and 30.2, lie above λ_+, and the observed λ_2 lies below both (Fig. 1c). In these nulls eigenvector 2 follows question difficulty: its loadings correlate with the questions' pass rates at a median $|r|$ of 0.62 and 0.63 (observed: 0.57, just below the 2PL null's 5th percentile and inside the Rasch null's 5–95% range); in the rotation-0 3PL null they are not ($|r| = 0.09$). To see whether this verdict could miss a second ability, we simulate from the 2PL fit and add, on the 125 split questions from STEM subjects, a second ability independent of the first, with the first ability's spread and median slope. Rank 2 then exceeds the null's 99th percentile in 69% (2PL) and 78% (Rasch) of datasets, and the family-wise test rejects in 91% and 93%. At half that strength the family-wise test rejects in about 20%, against 7% with no second ability, so a weaker second ability is not excluded (App. F).

Ranks 3 and 4. Rank 3 lies at the 98.5th (2PL) and 97.0th (Rasch) percentile and rank 4 at the 99.1st and 98.6th, in 2000 null draws. Rank 4 thus exceeds the 2PL 99th percentile by 0.03, but lies 0.01 below it in the 1000 draws of Fig. 1c, so its marker there is open (App. F). The family-wise test, which pools ranks 2 to 4, rejects both nulls ($p_M = 0.026$ and 0.041; Table A5). The excess is marginal: under one-ability truth, with the validity screen included, the test rejects in up to 8.7% of datasets at the 5% level, and calibrated against those datasets the observed p_M are 0.017 to 0.057. Dropping all 217 questions with an unparseable reply, which leaves 236 of the 391 columns and so less power, ends the rejection (App. B).

Source. Keeping one model of each sibling group, which removes five models, ends the rejection in all 24 ways of choosing, while removing five other models at random keeps it in all 24 trials (App. F). Removing one model each of two pairs named after these results, Phi-3-mini/Phi-3.5-mini and Llama-3-8B/Llama-3.1-8B, raises p_M to 0.22–0.69 (2PL) and 0.32–0.38 (Rasch), above all 190 removals of two other models ($p_M \leq 0.133$). Removing Llama-3.2-1B alone also ends the rejection but raises p_M less (0.060 and 0.089), so part of the excess depends on the letter-biased model. Simple baselines flag the same models: letter shares flag Llama-3.2-1B among three letter-biased models, and correlations of 2PL residuals rank the two pairs first and second of 276 model pairs (App. F); the spectrum adds the test.

4. Discussion

The directions we find beyond ability come from a shared seed or prompt within a model, and from one model's letter preference across models. After rotation, the test detects only a marginal excess, carried mostly by two pairs of near-duplicate models; it cannot exclude a second ability half as strong as the first, so it does not show that the ensemble lacks shared knowledge gaps. Whether these pairs fail on the same questions through knowledge gaps from common training data, through contamination or through another shared cause, we did not test; architecture, family, scale, training data and instruction tuning vary together in the ensemble. Item-level factor and item response analyses (Kipnis et al., 2025; Sha & Zhao, 2026; Yao et al., 2026) and agreement on the same wrong answer (Kim et al., 2025) may contain the same letter confound; we have not tested them. Before an ensemble's errors are read as epistemic, its options should be rotated, a question scored correct only if it is correct in every rotation, and the remaining spectrum tested against valid one-ability nulls, with a check of whether near-duplicate models carry what remains. A cross-model epistemic term such as that of Hamidieh et al. (2026) should be checked the same way.

Letter preference and the two pairs are the largest structure beyond ability that we find, not necessarily the only one; we test correctness only, not log-probabilities or reasoning traces. Power for structure carried by a few models was not simulated. The within-model runs cover only instruction-tuned models of 1.7B to 8B parameters, and the ensemble 500 MMLU questions. Per-rank thresholds are not corrected for the several nulls and scores we tried (App. F).

References

Bai, Z. and Silverstein, J. W. *Spectral Analysis of Large Dimensional Random Matrices*. Springer Series in Statistics. Springer, New York, 2 edition, 2010. doi: 10.1007/978-1-4419-0661-8.

Bao, Z., Pan, G., and Zhou, W. Tracy–Widom law for the extreme eigenvalues of sample correlation matrices. *Electronic Journal of Probability*, 17:1–32, 2012. doi: 10.1214/EJP.v17-1962. arXiv:1110.5208.

Burnell, R., Hao, H., Conway, A. R. A., and Hernández-Orallo, J. Revealing the structure of language model capabilities. *arXiv preprint arXiv:2306.10062*, 2023.

Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., Hesse, C., and Schulman, J. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

Corvelo Benz, N. L., Tsirtsis, S., Straitouri, E., Chatzi, I., Artola Velasco, A., Thejaswi, S., and Gomez-Rodriguez, M. Evaluation of large language models via coupled token generation. In *Proceedings of the 29th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 300 of *PMLR*, pp. 2215–2223, 2026.

Ding, K. When LLMs agree, are they right? Auditing self-consistency and cross-model agreement as confidence signals. *arXiv preprint arXiv:2607.08065*, 2026.

Farquhar, S., Kossen, J., Kuhn, L., and Gal, Y. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017):625–630, 2024. doi: 10.1038/s41586-024-07421-0.

Fei, X., Zhang, Y., Carneiro, S. A., and Vazirgiannis, M. Every wrong answer counts: Option-level psychometrics for LLM multiple-choice benchmarks. *arXiv preprint arXiv:2608.02966*, 2026.

Ferguson, G. A. The factorial interpretation of test difficulty. *Psychometrika*, 6(5):323–329, 1941. doi: 10.1007/BF02288588.

Hamidieh, K., Thost, V., Gerych, W., Yurochkin, M., and Ghassemi, M. Complementing self-consistency with cross-model disagreement for uncertainty quantification. In *International Conference on Learning Representations*, 2026.

Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., and Steinhardt, J. Measuring massive multitask language understanding. In *International Conference on Learning Representations (ICLR)*, 2021.

Hüllermeier, E. and Waegeman, W. Aleatoric and epistemic uncertainty in machine learning: An introduction to concepts and methods. *Machine Learning*, 110(3):457–506, 2021. doi: 10.1007/s10994-021-05946-3.

Jiang, T. The limiting distributions of eigenvalues of sample correlation matrices. *Sankhyā*, 66(1):35–48, 2004.

Johnstone, I. M. On the distribution of the largest eigenvalue in principal components analysis. *Annals of Statistics*, 29(2): 295–327, 2001. doi: 10.1214/aos/1009210544.

Kendall, A. and Gal, Y. What uncertainties do we need in Bayesian deep learning for computer vision? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Kim, E. M., Garg, A., Peng, K., and Garg, N. Correlated errors in large language models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *PMLR*, pp. 30038–30066, 2025.

Kipnis, A., Voudouris, K., Schulze Buschoff, L. M., and Schulz, E. metabench – a sparse benchmark of reasoning and knowledge in large language models. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2407.12844.

Lakshminarayanan, B., Pritzel, A., and Blundell, C. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Liu, Y., Duan, H., Zhang, Y., Li, B., Zhang, S., Zhao, W., Yuan, Y., Wang, J., He, C., Liu, Z., Chen, K., and Lin, D. MMBench: Is your multi-modal model an all-around player? In *Computer Vision – ECCV 2024*, volume 15064 of *Lecture Notes in Computer Science*, pp. 216–233. Springer, 2024. doi: 10.1007/978-3-031-72658-3_13.

Lord, F. M. and Novick, M. R. *Statistical Theories of Mental Test Scores*. Addison-Wesley, Reading, MA, 1968. With contributions by Allan Birnbaum.

Maimon, A., Cohen, A. D. N., Vishne, G., Ravfogel, S., and Tsarfaty, R. From benchmarks to skills: Low-rank factors for LLM evaluation. *Transactions of the Association for Computational Linguistics*, 14:1639–1653, 2026. doi: 10.1162/tacl.a.745.

Manakul, P., Liusie, A., and Gales, M. J. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 9004–9017, 2023. doi: 10.18653/v1/2023.emnlp-main.557.

Marčenko, V. A. and Pastur, L. A. Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR-Sbornik*, 1(4):457–483, 1967. doi: 10.1070/SM1967v001n04ABEH001994.

McDonald, R. P. and Ahlawat, K. S. Difficulty factors in binary data. *British Journal of Mathematical and Statistical Psychology*, 27(1):82–99, 1974. doi: 10.1111/j.2044-8317.1974.tb00530.x.

Min, S., Lyu, X., Holtzman, A., Artetxe, M., Lewis, M., Hajishirzi, H., and Zettlemoyer, L. Rethinking the role of demonstrations: What makes in-context learning work? In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pp. 11048–11064, 2022. doi: 10.18653/v1/2022.emnlp-main.759.

Pezeshkpour, P. and Hruschka, E. Large language models sensitivity to the order of options in multiple-choice questions. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pp. 2006–2017, 2024. doi: 10.18653/v1/2024.findings-naacl.130.

Pillai, N. S. and Yin, J. Edge universality of correlation matrices. *Annals of Statistics*, 40(3):1737–1763, 2012. doi: 10.1214/12-AOS1022. arXiv:1112.2381.

Rasch, G. *Probabilistic Models for Some Intelligence and Attainment Tests*. Danish Institute for Educational Research, Copenhagen, 1960.

Ravfogel, S., Svete, A., Snæbjarnarson, V., and Cotterell, R. Gumbel counterfactual generation from language models. In *International Conference on Learning Representations*, 2025.

Ruan, Y., Maddison, C. J., and Hashimoto, T. Observational scaling laws and the predictability of language model performance. In *Advances in Neural Information Processing Systems*, 2024.

Sha, T. and Zhao, S. BenchScope: How many independent signals does your benchmark provide? *arXiv preprint arXiv:2603.29357*, 2026.

Wang, X., Wei, J., Schuurmans, D., Le, Q., Chi, E., Narang, S., Chowdhery, A., and Zhou, D. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*, 2023.

Yao, L. H., Jarvis, N., Zhan, T., Ghosh, S., Liu, L., and Jiang, T. A geometric lens on LLM abilities through joint embedding item response theory. *Transactions on Machine Learning Research*, 2026. URL https://openreview.net/forum?id=VQe6p9wn5g. arXiv:2509.22888.

Zellers, R., Holtzman, A., Bisk, Y., Farhadi, A., and Choi, Y. HellaSwag: Can a machine really finish your sentence? In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 4791–4800, 2019. doi: 10.18653/v1/P19-1472.

Zhao, Z., Wallace, E., Feng, S., Klein, D., and Singh, S. Calibrate before use: Improving few-shot performance of language models. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, volume 139 of *PMLR*, pp. 12697–12706, 2021.

Zheng, C., Zhou, H., Meng, F., Zhou, J., and Huang, M. Large language models are not robust multiple choice selectors. In *International Conference on Learning Representations*, 2024.

A. Background

Positional preference in response models. Ding (2026) show with an option-shuffle control that part of a model's self-agreement is positional. In the option-level response model of Fei et al. (2026), a per-model positional preference barely helps to predict held-out correctness (accuracy 0.840 with it, 0.838 without). In a correctness matrix, such a preference can still create a second direction (Section 3.2).

Random-matrix tests for the number of factors. A factor shared by many variables lifts an eigenvalue of their sample correlation matrix beyond the Marchenko–Pastur edge λ_+ of Section 2: n variables with pairwise correlation κ have a population eigenvalue of $1 + (n-1)\kappa$. This eigenvalue grows with n, so an eigenvalue test can detect a correlation smaller than the sampling noise of one pairwise correlation, about $1/\sqrt{K}$. The MP law holds for the sample covariance matrices of independent entries with finite variance (Marčenko & Pastur, 1967; Bai & Silverstein, 2010) and, for i.i.d. entries, for their sample correlation matrices (Jiang, 2004). Johnstone (2001) showed for Gaussian data that the largest eigenvalue, minus the edge and divided by $\sigma_{\mathrm{TW}} = (1+\sqrt{\gamma})(1+1/\sqrt{\gamma})^{1/3}K^{-2/3}$, converges to the Tracy–Widom distribution. For sample correlation matrices, Bao et al. (2012) proved this limit for entries with a symmetric distribution and no centering, with fewer variables than observations, and with centering only for Gaussian entries; Pillai & Yin (2012) removed the symmetry condition and allowed more variables than observations, still without centering. Our $\mathbf{R}$ is centered at the sample means of Bernoulli columns with unequal pass rates; no theorem we know covers sample-mean centering of non-Gaussian entries, so the Monte Carlo null, not the limit, calibrates z.

Item response models. Item response models come from educational testing, where each test-taker has an ability and each question a difficulty; here the test-takers are language models. The 3PL form used here is $0.25 + 0.75\,\sigma(a_i(\theta_m - b_i))$ for four options. All are one-factor models: ability is the only source of dependence between questions.

B. Data collection

Questions. MMLU: the 50 subjects are the first in alphabetical order, and the ten questions of each are drawn at random. HellaSwag: 200 validation items, a random sample stratified by activity label. GSM8K: the first 100 test problems. The shared-prompt exemplars come from the 285 MMLU development questions: pass k uses five drawn at random from all subjects, independently of the other passes, so two passes can share an exemplar. The set is the same for every question of the pass and in both shared-prompt runs.

Prompts and decoding. Every model receives one user message rendered with its own chat template; Phi-2, which has none, receives the raw text. Prompts are tokenized with each model's own tokenizer and checked against an independent reference tokenization. In shared-prompt runs, the five exemplars, each followed by its answer letter, sit in this message between the instruction and the question. Generation uses vLLM 0.23.0 in bfloat16 with temperature T, top-$p = 1$, top-k disabled, min-$p = 0$, repetition penalty 1.0, no presence or frequency penalty, and at most 8 new tokens for MMLU and HellaSwag and 256 (or 1024) for GSM8K. Greedy decoding is $T = 0$. In independent-draw and shared-prompt runs, every sample (k, i) has its own seed, distinct from every other seed of the run; in shared-seed runs, all questions of pass k share one seed, and the seed changes from pass to pass. vLLM gives each seeded request its own random-number generator, so a request's sampling noise depends on its seed, not on its position in a batch, and requests with different seeds do not share noise (App. E).

Rotations and replicates. The 24 models answer every question in all four rotations, with deterministic kernel settings. GPU kernels can still differ between engine starts, so the ensemble was collected twice, with separate engine starts (replicates A and B). The paper reports replicate A.

Scoring. A multiple-choice reply is read as letter X when it begins with X, after optional "Answer:", parenthesis or bold markers, followed by punctuation, a line end, or words that do not negate or hedge the letter; or when its first line says "the answer is X" or a similar phrase. A reply that names several letters, negates its letter, or cannot be read this way is unparseable and scored wrong. The GSM8K prompt asks for the final answer after "####"; a reply is correct when the number after its last "####", or else its last number, equals the gold answer.

Unparseable replies. In the sampled runs at most 1.65% of replies are unparseable (Llama-3-8B on HellaSwag), and the shared-seed run and Qwen2.5-7B's shared-prompt run have none. In the ensemble, Mistral-7B v0.2 has 12.4% and every other model at most 9.6%. We either drop every question with an unparseable reply or replace each unparseable reply by the mean correctness of the question's parseable replies. Either way, every count of the independent-draw runs stays at zero, and the MP and buffered counts of Llama-3-8B's shared-prompt run stay at two. Across models, we rerun the valid nulls on both variants (Table A5). On all-4-correct, the family-wise rejection survives imputation and dropping the 95 questions whose all-4-correct score depends on an unparseable reply; dropping all 217 questions with any unparseable reply ends it (Section 3.3). In rotation 0, imputation leaves rank 2 above the letter-aware (App. F) and 3PL 99th percentiles; dropping the 112 affected questions moves it to the 98.3rd percentile of the letter-aware null, while it still exceeds the 3PL and the family-wise test rejects under both ($p = 0.013$ and 0.004).

Truncation on GSM8K. At 256 new tokens, 55% of Qwen2.5-7B's chain-of-thought replies reach the limit. Most of them stop before the final answer, and the scorer then reads the last number of an unfinished calculation: 13.6% of truncated replies are correct, against 96.7% of complete ones. Accuracy is 51.0% at 256 tokens and 90.3% at 1024 tokens. The run still tests independence: truncation changes which answers are correct, not whether two questions of a pass share a cause.

C. Test statistics and checks

Assumption C.1 (Independent draws). Given the model and the prompts, the entries c_{ki} are Bernoulli variables with $\Pr(c_{ki} = 1) = p_i$, independent across questions i and passes k.

Table A1. The test on every within-model run ($\varepsilon = 0.05$). λ_+: MP edge; TW z: Tracy–Widom statistic. Counts: MP (above λ_+) / buffered (above $1.05\,\lambda_+$, a margin for the fluctuation of $\lambda_{\max}$). GSM8K runs use $T = 1$.

Run	N_b	λ_+	$\lambda_{\max}$	TW z	Counts
Independent, MMLU ($K = 100, T = 1$)					
Qwen2.5-7B	43	2.74	2.56	-1.72	0/0
Mistral-7B-v0.3	83	3.65	3.45	-1.79	0/0
Llama-3-8B	173	5.36	4.97	-3.05	0/0
SmolLM2-1.7B	406	9.09	8.74	-2.17	0/0
Phi-3-mini	271	7.00	6.91	-0.65	0/0
Independent, HellaSwag ($K = 50, T = 1$)					
Qwen2.5-7B	16	2.45	2.19	-1.60	0/0
Mistral-7B-v0.3	65	4.58	4.05	-2.71	0/0
Llama-3-8B	86	5.34	4.75	-2.90	0/0
Independent, Qwen2.5-7B ($K = 30$)					
GSM8K, 256 tok.	62	5.94	5.32	-2.06	0/0
GSM8K, 1024 tok.	19	3.22	2.50	-2.97	0/0
MMLU, $T = 0.5$	25	3.66	3.39	-1.08	0/0
MMLU, $T = 1.5$	85	7.20	6.53	-2.07	0/0
Shared component, MMLU ($K = 100, T = 1$)					
Qwen, seed	41	2.69	10.00	$+70.2$	4/4
Qwen, prompt	93	3.86	4.77	$+7.9$	2/2
Llama, prompt	264	6.89	11.31	$+30.9$	2/2

Under Assumption C.1 the population correlation between two borderline questions is zero, so the population correlation matrix of the borderline columns is the identity. The filter keeps or drops each question on the basis of its own column alone, so the kept columns stay independent of each other. If every p_i stays a fixed distance from ε and $1-\varepsilon$, the kept set also equals the set of questions with $\varepsilon < p_i < 1-\varepsilon$ with probability tending to 1 as K and N grow in proportion, because the probability that f_i falls on the wrong side of a threshold decreases exponentially in K. No theorem we know gives the limit of z for $\mathbf{R}$ (App. A). At $K \le 100$ the distribution of z is shifted from the Tracy–Widom law: every z of the 12 independent-draw runs is negative, and their Monte Carlo nulls of z have means between -1.98 and -1.47. A z above every draw of its Monte Carlo null rejects Assumption C.1, as in the three shared-component runs, but does not say whether questions within a pass or the passes themselves are dependent.

Checks of independence. Three further checks on the independent-draw runs agree with Assumption C.1. A component that makes the questions of a pass right or wrong together would make the number of borderline questions a pass answers correctly vary more than independent draws at the pass rates predict. The ratio of the observed to the predicted variance of this number lies between 0.61 and 1.09 in the 12 runs, and no run is significantly overdispersed against a column-shuffle null (smallest $p = 0.27$); the two lowest ratios have one-sided underdispersion $p = 0.039$ and 0.045, neither significant after a Holm correction over the

48 tests (12 runs $\times$ 4 statistics). This check cannot detect a coupling that cancels in the number correct, such as that of the shared seed (App. E). Second, in the $T = 1$ MMLU runs of Qwen2.5-7B and Llama-3-8B, borderline questions of the same subject are no more correlated than other pairs (mean same-subject correlation -0.018, $p = 0.80$, and 0.002, $p = 0.37$). Third, a pairwise test that compares the largest Cramér's V and mutual information over all pairs of borderline questions with their permutation null finds no coupled pair, in correctness in all five MMLU runs at $T = 1$, and in the answer letters of Qwen2.5-7B (largest V 0.41 against a null 99th percentile of 0.72).

Borderline threshold. The threshold ε decides which questions enter the test, and with them N_b, λ_+ and the power. We repeat the test at $\varepsilon = 0.01$, 0.02, 0.05, 0.10, 0.15 and 0.20, with the filter re-applied at each threshold to the data and to every matrix of the run's Monte Carlo null (1000 matrices). Over all 12 independent-draw runs and six thresholds, no eigenvalue exceeds its edge, and every observed z lies below the 95th percentile of its null. The three shared-component runs keep at least two eigenvalues above their edge at every threshold. The six tests of a run are nested, not independent: raising ε removes the questions whose pass rates lie closest to 0 or 1. At $\varepsilon = 0.20$ the three smallest sets have 6, 7 and 10 questions (GSM8K at 1024 tokens, HellaSwag Qwen2.5-7B, MMLU at $T = 0.5$).

D. Power of the within-model test

We simulate correctness matrices at the pass rates of two independent-draw MMLU runs, Qwen2.5-7B ($N_b = 43$) and Llama-3-8B ($N_b = 173$), with a planted factor that makes some questions of a pass right or wrong together, and measure how often each count detects it. The factor loads on all of a run's borderline questions (diffuse), or on 30 or 9 of them chosen at random in each simulated matrix. Each entry comes from a hidden standard normal score: $c_{ki} = 1$ when the score falls below $\Phi^{-1}(f_i)$, with Φ the standard normal distribution function, so question i is correct with probability f_i. For a loaded question the score is $\sqrt{\rho}\,u_k + \sqrt{1-\rho}\,e_{ki}$, where the factor u_k is one standard normal draw shared by the loaded questions of pass k and e_{ki} is independent standard normal noise; for the other questions the score is e_{ki}. The score keeps unit variance, so each question keeps its pass rate, and the scores of two loaded questions correlate at ρ. Their 0/1 entries correlate less: averaged over pairs of borderline questions, about half of ρ at these pass rates.

Each simulated matrix has all 500 questions and $K = 100$ passes and goes through the test of Section 2, with the borderline filter re-applied. A count detects the factor when it is at least one; the detection rate is the power for $\rho > 0$ and the false-positive rate for $\rho = 0$. Each scenario has 500 simulated matrices, and $\rho = 0$ has 2000.

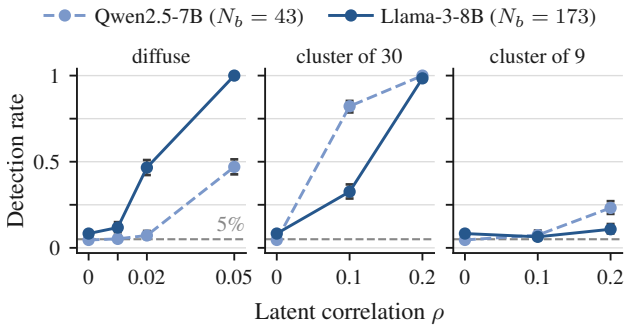


Figure A1. Detection rate of the MP count against the latent correlation ρ of a factor planted on all borderline questions or on 30 or 9 of them, at the pass rates of two MMLU runs. Whiskers: Wilson 95% intervals.

Power for a diffuse factor is higher at Llama-3-8B's 173 borderline questions, because the factor then loads four times as many questions; power for a fixed-size cluster is lower there, because the cluster is a smaller share of the questions (Fig. A1): a 9-question cluster at $\rho = 0.2$ is detected in 23% of Qwen2.5-7B and 11% of Llama-3-8B simulations. The buffered count trades power for its lower false-positive rate (diffuse factor at $\rho = 0.05$: 22% at Qwen2.5-7B, 99.8% at Llama-3-8B). At $\rho = 0$ the false-positive rates are 4.6% and 8.3% for the MP count and 0.65% and 0.35% for the buffered count. Runs with other numbers of borderline questions were not simulated; we expect power for a diffuse factor to fall, and for a fixed cluster to rise, as N_b shrinks.

E. What the shared component couples

Shared noise under a shared seed. At each generation step, the random-number generator of a seeded request draws one exponential random number, the noise, for every token of the vocabulary, and the request emits the token whose probability at temperature T, divided by its noise, is largest. This is Gumbel-max sampling, since the negative logarithm of an exponential random number is a Gumbel random number: each answer, taken alone, follows the model's distribution for its question. Two requests with the same seed draw the same noise vector at every step, whatever their prompts; coupled generation and counterfactual sampling use this sharing on purpose (Corvelo Benz et al., 2026; Ravfogel et al., 2025). If a shared-seed pass draws a small noise value for the token "B", the sampler divides the probability of "B" by that small value in every question of the pass whose letter is generated at that step, and so favors "B" in all of them at once.

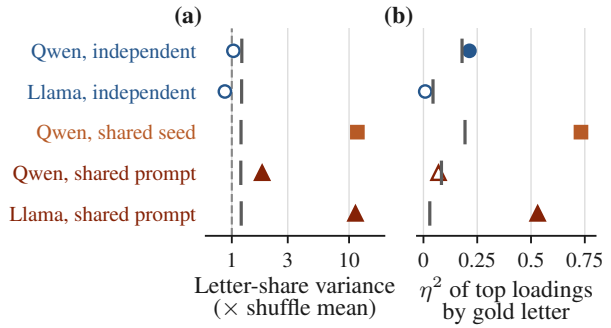


Figure A2. A shared seed or prompt makes a pass's letter shares vary beyond chance; of the shared runs, only the seed and Llama-3-8B's prompt also group the top loadings by gold letter. **(a)** Across-pass variance of a pass's letter shares, summed over letters, over its mean when answers are shuffled across passes. **(b)** η^2 of the top eigenvector's loadings by gold letter. Grey tick: null 99th (a) or 95th (b) percentile; filled: beyond it.

What the directions follow. With a shared seed, the across-pass variance of the fraction of a pass's answers that are each letter, summed over letters, is 11.8 times its mean when each question's answers are shuffled across passes (1.03 for the independent-draw run; Fig. A2a), and the top eigenvector's loadings group by gold letter (Sec. 3.1). A pass whose noise favors one letter gains on questions with that gold letter and loses on the others, so the coupling cancels in the number correct per pass (observed over predicted variance 0.83). In its shared-prompt run, Llama-3-8B copies the exemplars' letters, as few-shot prompts are known to pull answers toward their labels (Zhao et al., 2021). Let r be the correlation, over passes, between the number of a pass's five exemplars whose answer is letter L and the pass's share of answers L: it lies between 0.35 and 0.47 for the four letters, the letter shares vary 11.3 times their shuffle mean, and the number correct per pass is overdispersed (ratio 1.51, $p = 0.001$). Neither direction groups by subject beyond chance ($p = 0.23$ and 0.35). Qwen2.5-7B's shared-prompt run behaves differently: its letter shares vary 1.8 times their shuffle mean, but they do not follow the exemplars' letters (r between -0.09 and 0.09), and its loadings do not group by gold letter ($\eta^2 = 0.07$). The exemplar set still matters: two passes that share an exemplar answer the borderline questions more alike than two that share none, by 10.7 null standard deviations when the exemplar sets are permuted across passes (5.9 for Llama-3-8B; permutation $p = 0.0005$ for both). Its loadings group weakly by subject ($p = 0.042$, uncorrected), and a same-subject exemplar raises accuracy on borderline questions by 3.2 points ($p = 0.033$); only these two effects link the coupling to the exemplars' content. The Qwen2.5-7B independent-draw run shows $\eta^2 = 0.21$ ($p = 0.023$; Fig. A2b) although its top eigenvalue lies below its edge; we do not interpret it. The answer-letter test of App. C finds no pair of its borderline questions with letters associated beyond chance, but that test looks for single coupled pairs, not for a grouping of the loadings by gold letter.

F. The 24-model ensemble

Roster. Five Qwen (Qwen2.5 at 0.5B, 1.5B, 3B and 7B, and Qwen2-7B), five Mistral (Mistral-7B v0.2 and v0.3, Zephyr-7B-beta, Ministral-8B, Mistral-Small-22B), four Llama (Llama-3.2 at 1B and 3B, Llama-3-8B, Llama-3.1-8B), three Phi (Phi-2, Phi-3-mini, Phi-3.5-mini), two Gemma-2 (2B and 9B), and one each of SmolLM2-1.7B, OLMo-2-7B, Yi-1.5-6B, DeepSeek-LLM-7B and Granite-3.1-8B: ten families, five with a single model. Phi-2 is a base model; the other 23 are instruction-tuned. Greedy accuracy in rotation 0 runs from 35.0% (Llama-3.2-1B) to 72.8% (Gemma-2-9B).

One-ability nulls. Every null is simulated conditional on every kept column staying split, so each null matrix has the observed number of columns and the observed λ_+; each has 1000 draws. Rasch is fitted by joint maximum likelihood. The fixed-margin null draws uniformly among 0/1 matrices with the observed row and column sums. The 2PL treats the abilities as fixed and integrates the question parameters over a normal population (marginal maximum likelihood); it is fitted to all 500 columns and simulated on the split ones. The Bock–Aitkin 2PL integrates the abilities instead, with slopes bounded by 4. The 3PL uses the floor 0.25 for every score. The letter-aware null is a know-or-guess model fitted to rotation 0, with each model's letter preference π_m estimated from all four rotations. The 4-rotation letter know-or-guess (KG-l) null is fitted like KG-u (Section 2), with guessing by π_m; both simulate every rotation.

Validity. The 2PL, Rasch and fixed-margin nulls put the observed rotation-0 λ_1 above all their draws, and the 4-rotation know-or-guess and Bock–Aitkin nulls put the all-4-correct λ_1 below all theirs. A null counts only if the observed λ_1 lies between its 1st and 99th percentiles, for every score and every null, the soft-preference null included. The rule was set after the first analyses; the size simulation below tests it.

Construction family. All-4-correct is one of several ways to combine a model's four rotations, and a letter preference that acts on choices the model partly knows can leak into any of them. Table A3 gives each score's directions beyond ability under the nulls valid for it. In rotation 0 without Llama-3.2-1B (Section 3.2), the letter-aware null also rejects ($p_M = 0.015$), the letter grouping moves to eigenvector 3 ($\eta^2 = 0.21$), and ranks 3 and 4 lie at the 98.5th to 99.9th

Table A2. Percentile of the observed λ_1 among each one-ability null's draws; bold: valid (1 to 99). Draws per null: 1000 (rotation 0), 2000 (all-4-correct). –: fitted to rotation 0 only.

Null	Rotation 0	All-4-correct
2PL	100	**31.5**
Rasch	100	**54.2**
3PL	**76.4**	100
Letter-aware	**86.7**	–
Uniform know-or-guess	**98.7**	0.0
Soft preference (App. F)	**85.7**	0.0

percentiles of the 3PL and letter-aware nulls; in their 1000 draws, rank 4 exceeds the 99th percentiles by 0.35 and 0.57. On ≥ 2 of 4 the family-wise test agrees with the per-rank rule ($p_M = 0.002$ under the 2PL, 0.86 under the 3PL). On ≥ 3 of 4 it rejects the Rasch, KG-l and 2PL nulls ($p_M = 0.008$, 0.007 and 0.009) but not KG-u (0.28), and in its 2000 draws the per-rank counts of the Rasch and KG-l nulls differ from the table (1 and 3), because these ranks lie close to their 99th percentiles. On ≥ 1 of 4 only the SP null is valid, and the family-wise test also finds nothing ($p_M = 0.996$); the mean has no valid null. Removing the 23 questions whose options refer to other options ("all of the above", "A and B only") leaves the counts of rotation 0 and all-4-correct unchanged, lowers those of ≥ 2 (2PL 2) and ≥ 3 (Rasch 1, KG-l 0), and makes the BA null valid on ≥ 1 of 4, with no direction.

Nulls with a letter preference. In rotation 0, rank 2, and no other rank, also exceeds the valid letter-aware null, and the family-wise test rejects it ($p_M = 0.002$). This null lets the preference act only when a model guesses. When the preference also tilts choices that a model partly knows, as in the soft-preference null below, rank 2 exceeds the letter-aware null in 60% and the 3PL null in 99.5% of 200 datasets simulated from it with no second factor (each null refitted, 200 draws).

Soft-preference null. In the SP null, model m chooses option j of question i in a rotation with probability proportional to $\exp\big(b_{ij} + \mathbb{1}[j = g_i](a_i\theta_m + e_{mi})\big)\,\pi_m(L_j)$, where g_i is the gold option, $b_{ig_i} = 0$, L_j is the letter at which the rotation shows option j, and $e_{mi} \sim N(0, \tau^2)$ is shared by the four rotations of a cell and independent across models. Ability θ_m is the only link between questions, and the letter preference acts on every choice, also when the model partly knows the answer. The fit is penalized maximum likelihood over the parsed choices of all four rotations ($\tau = 1.17$). Simulations keep the observed unparseable cells, and the rotation-0 result depends on this: without the mask the observed λ_2 lies at the 99.8th percentile of the null, with it at the 95.1st. Refitting inflates the slopes, so we calibrate: 200 datasets simulated from the fit, each refitted with its own null. The calibrated p of rank j is the fraction of these datasets whose λ_j percentile is at least the observed one. Each refitted null uses 1000 draws. On rotation 0 the SP null is valid (Table A2); λ_2 lies at its 95.1st percentile, and its calibrated p (Section 3.2) is 0.015 by the Monte Carlo convention $(1 + k)/(1 + n)$, with k of the $n = 200$ datasets at least as extreme; without Llama-3.2-1B the null is invalid. On ≥ 3 of 4 and all-4-correct it is invalid (λ_1 at the 0th percentile), so its rejections there are not evidence.

Predictive check. The soft-preference model fails its preset predictive check, which compares 500 datasets simulated from the fit with the data. Per-model accuracy and letter shares fit, but the number of rotations, 0 to 4, in which a model answers a question correctly is covered for 82.5% of model-by-count cells, below the required 90%. The weakest, letter-biased models have too many questions correct in exactly one rotation, as expected if they choose one letter almost deterministically, and stronger models too many correct in all four (pooled: 0.196 observed against 0.177 simulated for one rotation, 0.382 against 0.366 for four). The calibration corrects the bias of the fitting pipeline, not this misfit, so the soft-preference verdicts are not fully validated.

Family-wise test. The test uses new null draws, with seeds disjoint from those of all earlier runs. The statistic M is the largest of the percentiles of λ_2, λ_3 and λ_4 among 2000 new null draws (ties count half); each draw's own M_d is computed against the other draws, and $p_M = (1 + \#\{d : M_d \geq M\})/2001$, at level 0.05 (Table A5). On 200 datasets simulated from the SP fit with no second factor (500 null draws each), the refitted 2PL null on ≥ 2 and ≥ 3 of 4 is valid in 98.5% and 100% of the datasets, and it is valid and rejects in 98.5% and 58% of all datasets. The refitted Rasch null is valid in 0% and 18% of the datasets; it is valid and rejects in 0% and 11% of all datasets. The SP fit is valid on ≥ 2 but not on ≥ 3, so the ≥ 3 rates are indicative only. No simulation covers the KG-l test; we draw no conclusion from the ≥ 3 rejections. Three null draws reach $M = 100$, so p_M cannot fall below 0.002, the value in rotation 0 and on ≥ 2 of 4.

Size. We simulate 300 all-4-correct datasets from each fitted one-ability model, the 2PL and Rasch, with the nulls' own simulator, refit both nulls to each dataset (500 draws each), and apply the validity rule and the family-wise test; Table A5 gives the rejection rates and the observed p_M calibrated against them. The 2PL test rejects in 6.7% of datasets under 2PL truth and 8.7% under Rasch truth (95% intervals 4.1–10.1% and 5.7–12.4%), so it is not size-controlled by the bound of 7.5% we set before the run. The Rasch test rejects rarely under Rasch truth (0.7%), mainly through the validity screen: the refitted Rasch null is valid in only 23%

Table A3. Construction family and soft-preference (SP) null. Each score combines a model's four rotations: correct in rotation 0, in at least k of 4, or the mean. Valid nulls, each with its number of directions beyond ability by the per-rank rule: ranks $2, 3, \ldots$ that each exceed the null's 99th percentile, counted until one does not. Results for all 24 models and without Llama-3.2-1B (w/o); 1000 draws per null; "none": no valid null other than SP. LA: letter-aware; KG-l, KG-u: 4-rotation letter and uniform know-or-guess; BA: Bock–Aitkin 2PL. SP: rank-2 calibrated p; "inv.": invalid.

Score	Cols	λ_1–λ_4	Valid nulls: ranks beyond ability		SP: rank-2 cal. p
			24 models	w/o Llama-3.2-1B	24 / w/o
Rotation 0	460	85.9, 49.9, 29.7, 27.3	3PL 1, KG-u 1, LA 1	3PL, LA 0	0.010 / inv.
≥ 1 of 4	334	34.2, 27.2, 23.7, 21.4	none	BA 0	0.750 / inv.
≥ 2 of 4	444	108.3, 42.3, 30.2, 23.0	2PL 3, 3PL 0	2PL 1, 3PL 0	0.245 / 0.280
≥ 3 of 4	439	103.4, 37.8, 25.5, 22.1	Rasch 2, KG-l 2; 2PL, KG-u 0	2PL, KG-l, KG-u 0	inv. / 0.285
All 4	391	92.1, 29.0, 23.3, 20.8	Rasch, 2PL 0	Rasch, 2PL, KG-u 0	inv. / inv.
Mean	489	127.5, 43.1, 28.5, 25.7	none	none	inv. / inv.

of those datasets. The validity rule is not sufficient: it does not detect wrong slopes (the Rasch null is valid in every dataset simulated from the 2PL fit), and it rejects the correct Rasch model, refitted by joint maximum likelihood, in most datasets simulated from it.

Sibling models. The sibling groups (Section 2) were chosen from model metadata. Models of one family at different sizes are not siblings. Each of the $2 \cdot 2 \cdot 3 \cdot 2 = 24$ subsets that keep one model per group has 19 models; both nulls are refitted to each, with 1000 draws. None of the 24 subsets rejects (Table A5). The control removes five of the 15 models outside the groups, chosen at random, 24 times, so that 19 models remain as well; all 24 reject. The subsets of each kind overlap heavily, so the comparison is one contrast, not 24 replicates. The random removals include models that share a developer with the groups (Qwen2.5 at 0.5B to 3B, Llama-3.2, Ministral-8B and Mistral-Small), and they are not matched to the 7B–8B chat models of the groups. On all-4-correct both nulls are valid in every sibling and random subset. In rotation 0 the 2PL and Rasch nulls are valid in none of the 24 sibling subsets and the uniform know-or-guess null in 17; the 3PL and letter-aware nulls are valid and reject in all 24, so the letter direction does not depend on the siblings.

Pairs and simple baselines. The pair analysis compares the four subsets that remove one model of each of the two pairs of Section 3.3 with all 190 subsets that remove two of the other 20 models (1000 draws per subset; every null valid in every subset). All four pair removals lie above every reference removal under both nulls (Table A5); 90% (2PL) and 74% (Rasch) of the reference removals still reject. Removing Qwen2-7B or Qwen2.5-7B makes the rejection stronger (mean reference p_M 0.010 and 0.006 under the 2PL), so the Qwen pair is not part of the excess. The effect is specific to the two pairs: removing the Mistral-7B v0.2 and v0.3 near-duplicates (residual rank 3 in Table A4) keeps the rejection ($p_M = 0.019$ under both nulls), as do the other two removals within the Mistral group (p_M 0.004 to 0.023). Nor does removing two strong chat models explain it: the 15 reference removals whose two models have rotation-0 accuracy between 0.62 and 0.71 (the pair models: 0.644 to 0.684; no two other models lie inside that narrower range) give p_M at most 0.052 (2PL) and 0.113 (Rasch), below every pair removal. The letter-biased model contributes too. Without Llama-3.2-1B alone (Section 3.3), the uniform know-or-guess null becomes valid and gives $p_M = 0.15$; a second statistic fixed with the test but not used for decisions, $\lambda_2 + \lambda_3 + \lambda_4$ against its null draws, still gives $p = 0.023$ (2PL) and 0.016 (Rasch). Of the 19 reference removals that include Llama-3.2-1B, 12 (2PL) and 6 (Rasch) still reject, with p_M at most 0.099 and 0.130. Removing Llama-3.2-1B together with one model of each pair gives 0.30 to 0.59 and 0.38 to 0.60 (1000 draws), close to the pair removals alone. Table A4 gives two simple baselines. Letter shares in rotation 0 flag the three models whose most frequent letter exceeds half of their replies; these shares count unparseable replies in the denominator, whereas the preference of 0.90 for A in Section 3.2 is estimated over guesses. The correlations of model pairs' residuals (observed minus 2PL-expected correctness over the 391 all-4-correct columns) rank the Phi and Llama pairs first and second of 276; the spectrum adds a test of whether such candidates exceed one ability.

Power. To measure what the all-4-correct verdict can exclude, we simulate from the 2PL fit to the all-4-correct matrix, which is valid, with a second ability added. Each kept column draws (a_i, b_i) from its posterior, as the 2PL null does, and on the 125 of the 391 questions from the 18 STEM subjects (mathematics, statistics, the natural sciences, computer science and electrical engineering) the logit gains $s\,\bar{a}\,\mathrm{sd}(\theta)\,z_m$, where $z_m \sim N(0,1)$ is independent of θ_m, $\bar{a} = 1.71$ is the median slope and $\mathrm{sd}(\theta) = 1$. At $s = 1$ the second ability has the spread and the median slope of the first. Both nulls are refitted to each of 200 datasets per s (500 draws each; rates in Table A5). At $s = 2$ both rules

Table A4. Simple baselines. Top: rotation-0 letter shares (unparseable replies in the denominator) of the models whose largest share exceeds 0.5; every other model has a largest share of at most 0.35. Bottom: correlation of model pairs' residuals after the all-4-correct 2PL fit, with rank among 276 pairs (mean -0.04, standard deviation 0.08).

Model	A	B	C	D
Llama-3.2-1B	0.81	0.04	0.09	0.05
Qwen2.5-0.5B	0.23	0.06	0.12	0.58
SmolLM2-1.7B	0.07	0.20	0.17	0.56

Model pair	r	**Rank**
Phi-3-mini, Phi-3.5-mini	0.40	1
Llama-3-8B, Llama-3.1-8B	0.27	2
Mistral-7B v0.2, v0.3	0.20	3
Mistral-7B v0.3, Llama-3-8B	0.19	4
Mistral-7B v0.2, Zephyr-7B	0.17	5
Qwen2-7B, Qwen2.5-7B	0.03	37

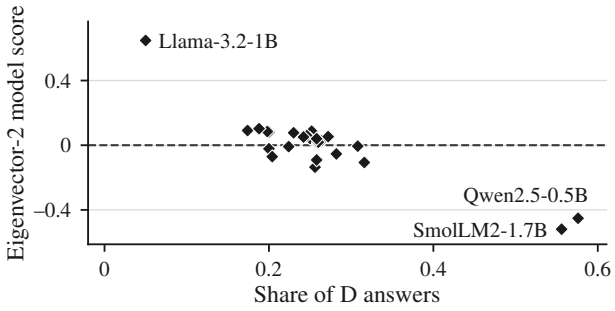


Figure A3. Each model's rotation-0 score on eigenvector 2 of the 24-model matrix against its share of D answers.

detect the second ability in at least 97% of datasets. The median λ_2 is 28.2 at $s = 0$, 28.3 at $s = 0.5$ and 33.8 at $s = 1$, against the observed 29.0.

Letter preference. In Figure A3, $r = -0.90$ has permutation $p = 10^{-4}$. Figure 1b uses 440, 433 and 448 questions in rotations 1–3. Eigenvector 1 also carries a letter part ($\eta^2 = 0.22$ by gold letter), and eigenvectors 3 and 4 group only slightly ($\eta^2 = 0.087$ and 0.061). In expectation, benchmark-level scores cannot carry a letter preference when gold letters are balanced, as nearly here (118, 124, 129, 129 for A to D).

The remaining eigenvectors. The all-4-correct eigenvector 2 groups its question loadings by subject ($\eta^2 = 0.21$, $p = 0.001$), but a cubic in question difficulty explains 39% of its loading variance, and after removing it the grouping is at chance ($p = 0.24$). Of the borderline directions of the other scores, only ≥ 2 ranks 2 and 3 group by subject after this removal ($p = 0.010$ and 0.034, uncorrected over eight eigenvectors). The models' scores on the all-4-correct eigenvector 4, below λ_+, group by family ($\eta^2 = 0.74$, $p = 0.006$); in rotation 0, after a Holm correction, only eigenvector 4 aligns with family ($p = 0.017$), and same-family model pairs correlate little more than other pairs (mean r 0.414 against 0.375, model-label permutation $p = 0.085$).

Table A5. Family-wise test on all-4-correct (reject at $p_M < 0.05$). Subsets: number that reject, and the range or maximum of p_M. Pair removals: one Phi and one Llama model; other two removed: the 190 removals of two other models (fraction that reject, maximum p_M). Unparseable: replies imputed, or the 95 questions whose score they affect, or all 217 questions with any, dropped (1000 draws). Size: rejection rate in 300 datasets simulated from each fitted one-ability model; calibrated p_M: under 2PL, Rasch truth. Power: family-wise / rank-2 rule, 200 datasets per strength s simulated from the 2PL fit.

	2PL	**Rasch**
Pct. of $\lambda_2, \lambda_3, \lambda_4$	71.8, 98.5, 99.1	94.6, 97.0, 98.6
p_M, 24 models	0.026	0.041
Sibling subsets	0/24 (0.09–0.65)	0/24 (0.12–0.69)
Random subsets	24/24 ($\leq$0.047)	24/24 ($\leq$0.046)
Pair removals	0/4 (0.22–0.69)	0/4 (0.32–0.38)
Other two removed	0.90 ($\leq$0.099)	0.74 ($\leq$0.133)
Unparseable: impute	0.019	0.042
drop 95	0.036	0.043
drop 217	0.23	0.12
Size, 2PL truth	0.067	0.067
Size, Rasch truth	0.087	0.007
Calibrated p_M	0.033, 0.047	0.057, 0.017
Power, $s = 0.5$	0.20 / 0.01	0.21 / 0.07
Power, $s = 1$	0.91 / 0.69	0.93 / 0.78

Replicates and near ties. Replicate B differs from A in 6 of 48,000 chosen options, so it tests GPU nondeterminism, not resampling; eigenvalues differ between the replicates by at most 0.081 over all matrices. In replicate B, λ_4 (20.82) lies above the 2PL rank-4 99th percentile of 1000 draws (20.78; Monte Carlo $p = 0.009$); in replicate A it lies 0.01 below (Section 3.3). A near tie is a cell whose two most probable first-token letters differ by under 0.13 nats in log probability in either replicate (5.0% of cells). Replacing near ties by the majority answer over rotations changes no verdict. Without the 297 questions with a rotation-0 near tie (203 remain, so less power), rank 2 exceeds the 3PL but not the letter-aware null (both valid, λ_1 near their 13th and 14th percentiles).